King Fahd University of Petroleum and Minerals
Department of Mathematics and Statistics

Math 201
Final Exam
T-092
Sunday 20/06/2010
Net Time Allowed: 180 minutes

MASTER VERSION

1. The cartesian equation for the curve $x = 3 + \cos t$, $y = 2 + \sin t$ is:

(a) $x^2 + y^2 - 6x - 4y = -12$

(b) $(x-3)^2 + (y-2)^2 = -1$

(c) $(x-3)^2 + (y-2)^2 = 4$

(d) $x^2 - y^2 - 6x + 4y = -4$

(e) $x^2 - y^2 = 4$

2. Find the area of the region bounded by $r = 1 + \sin\theta$ for $\frac{\pi}{4} \leq \theta \leq \pi$.

(a) $\frac{9\pi}{16} + \frac{\sqrt{2}}{2} + \frac{9}{8}$

(b) $\frac{9\pi}{16} + \frac{\sqrt{2}}{2}$

(c) $\frac{9\pi}{4}$

(d) $\frac{2\pi}{3}$

(e) $\frac{2\pi}{6}$

3. If $\vec{u} = 7\hat{\imath} + 3\hat{\jmath} + 5\hat{k}$ and $\vec{v} = -8\hat{\imath} + 4\hat{\jmath} + 2\hat{k}$, then $Comp_{\vec{v}}\vec{u}$ is:

(a) $-\dfrac{17}{\sqrt{21}}$

(b) $-\dfrac{17}{\sqrt{21}}\left(-8\hat{\imath} + 4\hat{\jmath} + 2\hat{k}\right)$

(c) $-\dfrac{34}{\sqrt{69}}$

(d) $-\dfrac{34}{\sqrt{80}}$

(e) $\dfrac{-14\hat{\imath} - 54\hat{\jmath} + 52\hat{k}}{\sqrt{21}\sqrt{83}}$

4. An equation of the plane that passes through the point $(2, -1, 3)$ and contains the line $\dfrac{x+1}{-3} = \dfrac{y-2}{2} = \dfrac{z-1}{-1}$ is:

(a) $x + 3y + 3z = 8$

(b) $x - 3y + 3z = 14$

(c) $x - 2y + 3z = 8$

(d) $x - 2y + z = -4$

(e) $2x - y + z = 1$

5. The quadric surface $x^2 - y^2 + z^2 - 4x - 2y - 2z + 4 = 0$ represents:

(a) a circular cone with vertex (2,-1,1) and axis parallel to y-axis.

(b) a circular cone with vertex (2,1,1) and axis parallel to y-axis.

(c) a paraboloid with vertex (2,-1,1)

(d) a hyperboloid with vertex (2,1,1)

(e) an ellipsoid with center (2,-1,1)

6. The limit $\lim_{(x,y)\to(0,0)} \left(x^2+y^2\right) \ln\left(x^2+y^2\right)$ is equal to:

(a) 0

(b) 1

(c) ∞

(d) -1

(e) 2

7. If $w = x^2 \cos(xy)$ then $\dfrac{\partial w}{\partial x}\left(\dfrac{1}{2}, \pi\right) + 2\pi \dfrac{\partial w}{\partial y}\left(\dfrac{1}{2}, \pi\right)$ is equal to:

(a) $-\dfrac{\pi}{2}$

(b) 0

(c) $\dfrac{\pi}{2}$

(d) $-\dfrac{1}{2}$

(e) $\dfrac{1}{2}$

8. Let $f(x, y) = \sqrt{20 - x^2 - 7y^2}$ and let $L(x, y)$ be the linearization of f at $(2, 1)$. Using $L(x, y)$, the value of $f(2.1, 1.2)$ is approximately equal to:

(a) 2.467

(b) 4.27

(c) 2.71

(d) 1.90

(e) 3.12

9. For the function z defined implicitly by the equation $xyz = \sin(e^{xyz} + z)$, the value of $\dfrac{\partial z}{\partial y}(0, -1)$ is:

(a) 0

(b) 1

(c) -1

(d) $\dfrac{1}{2}$

(e) $-\dfrac{1}{2}$

10. The points on the hyperboloid $x^2 - y^2 + 2z^2 = 1$ where the normal line is parallel to the line passing through the points $(3, -1, 0)$ and (5,3,6) are:

(a) $\left(\pm\sqrt{\dfrac{2}{3}}, \mp\dfrac{2\sqrt{2}}{\sqrt{3}}, \pm\sqrt{\dfrac{3}{2}}\right)$

(b) $\left(\pm\dfrac{3}{2}, \mp\dfrac{8}{3}, \pm\dfrac{3}{2}\right)$

(c) $\left(\pm\dfrac{1}{\sqrt{3}}, \mp\dfrac{1}{\sqrt{3}}, \pm\dfrac{1}{\sqrt{2}}\right)$

(d) $\left(1, -2, \dfrac{3}{2}\right)$

(e) $\left(\pm\sqrt{\dfrac{3}{2}}, \mp\sqrt{6}, \pm\dfrac{3\sqrt{3}}{2\sqrt{2}}\right)$

11. The point on the plane $x - y + z = 4$ that is closest to the point $(1, 2, 3)$ is:

(a) $\left(\frac{5}{3}, \frac{4}{3}, \frac{11}{3}\right)$

(b) $(5, 2, 1)$

(c) $\left(\frac{7}{2}, 2, \frac{5}{2}\right)$

(d) $(1, 2, 3)$

(e) $(0, 0, 0)$

12. The function $f(x, y) = 7x^2y + 10xy^2$ has:

(a) 1 critical point

(b) 2 critical points

(c) 3 critical points

(d) 4 critical points

(e) no critical point

13. The plane $x+2y+4z=4$ intersects the cylinder $x^2+y^2=5$ in an ellipse. If P is the highest point on the ellipse and Q is the lowest point on the ellipse, then $\frac{1}{2}(P+Q)$ is:

(a) $(0,0,1)$

(b) $(-1,-2,4)$

(c) $(1,2,4)$

(d) $(0,0,4)$

(e) $(0,0,0)$

14. An approximate value of the integral $\int\int_R \left(2\cos^2 y - \sin^2 x\right) dA$ where $R = \left[0, \frac{\pi}{2}\right] \times \left[-\frac{\pi}{6}, \frac{\pi}{6}\right]$, using $m=3, n=2$ and with upper left corner points is equal to:

(a) $\frac{17\pi^2}{72}$

(b) $-\frac{17\pi}{72}$

(c) $\frac{19\pi^2}{72}$

(d) $-\frac{20\pi}{3}$

(e) 6

15. The value of $\int\int_R \frac{1+x^2y}{1+y^2}dA$, where $R = [0,1] \times [0,1]$ is equal to:

(a) $\frac{\pi}{4} + \frac{\ln 2}{6}$

(b) $\frac{\pi}{2} + \frac{\ln 2}{3}$

(c) $\frac{\pi}{4} + \frac{\ln 2}{3}$

(d) $\frac{\pi}{2} + \frac{\ln 2}{6}$

(e) $\frac{\pi}{3} + \frac{\ln 2}{3}$

16. The value of $\int_0^1 \int_{3y}^3 e^{x^2} dxdy$ is :

(a) $\frac{e^9 - 1}{6}$

(b) $\frac{e^9}{6}$

(c) $\frac{2(e^9 - 1)}{6}$

(d) $e^9 - e^{\frac{9}{4}}$

(e) $e^9 - e^{\frac{9}{4}} + 2$

17. The value of the double integral $\int_0^4 \int_0^{\sqrt{16-x^2}} e^{-\left(x^2+y^2\right)} dydx$ is equal to:

(a) $\dfrac{\pi\left(1-e^{-16}\right)}{4}$

(b) $\dfrac{\pi\left(1+e^{-16}\right)}{4}$

(c) $\dfrac{\pi\left(1-e^{-16}\right)}{2}$

(d) $\dfrac{\pi\left(1+e^{16}\right)}{4}$

(e) $\dfrac{\pi\left(1-e^{16}\right)}{4}$

18. The value of the integral $\int\int_E \int (x+2y)\, dV$, where E is bounded by the the parabolic cylinder $y = x^2$ and the planes $y = x, z = 0$ and $z = x$ is:

(a) $\dfrac{2}{15}$

(b) $\dfrac{4}{15}$

(c) $\dfrac{13}{15}$

(d) $\dfrac{1}{15}$

(e) $\dfrac{7}{15}$

19. The value of $\int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}\int_{\sqrt{x^2+y^2}}^{2}(x^2+y^2)^{\frac{3}{2}}dzdydx$ is:

(a) $\dfrac{64\pi}{15}$

(b) $\dfrac{64\pi}{5}$

(c) $\dfrac{128\pi}{5}$

(d) $\dfrac{128\pi}{15}$

(e) $\dfrac{64}{15}$

20. The value of the integral $\int\int\limits_{E}\int \dfrac{e^{x^2+y^2+z^2}}{\sqrt{x^2+y^2+z^2}}dV$, where $E=\left\{(x,y,x)\,|x^2+y^2+z^2\leq 1\right\}$, is:

(a) $2\pi\,(e-1)$

(b) $2\pi e$

(c) $\pi\,(e-1)$

(d) πe

(e) $4\pi\,(e-1)$

Q	MM	V1	V2	V3	V4
1	a	c	a	c	c
2	a	b	e	a	e
3	a	e	d	c	d
4	a	b	b	d	c
5	a	c	c	e	d
6	a	e	d	b	c
7	a	c	a	c	c
8	a	e	d	a	e
9	a	e	d	c	d
10	a	a	b	e	d
11	a	d	a	c	e
12	a	e	e	c	e
13	a	b	c	d	c
14	a	c	b	d	c
15	a	c	b	b	d
16	a	d	c	c	c
17	a	c	e	b	d
18	a	c	e	d	c
19	a	e	a	c	e
20	a	c	b	b	a